Local LLM Agent for International Student Support

Concept Proposal for Project B – Business Analytics

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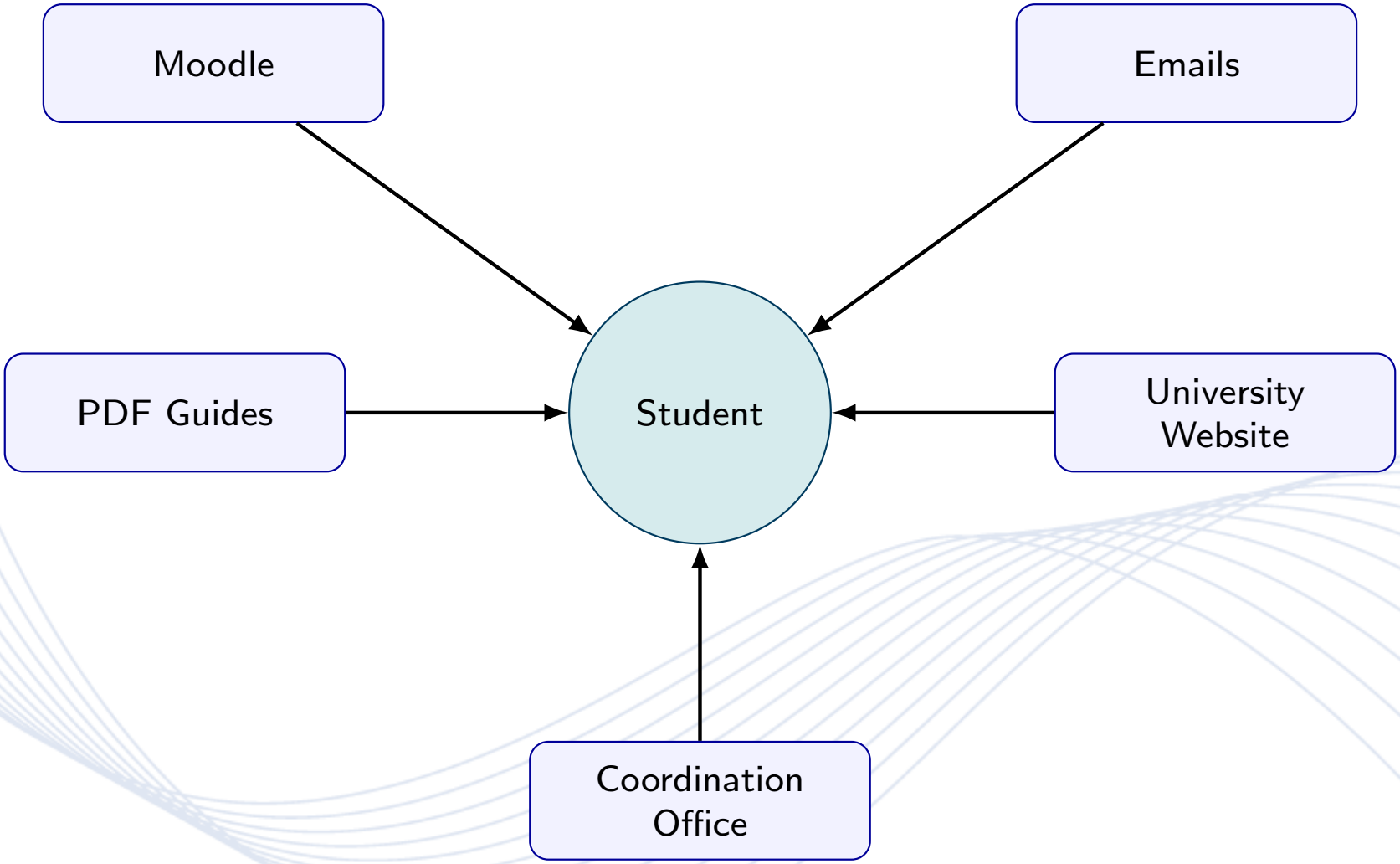
Motivation and Problem

Motivation

- International students must quickly understand many university systems and procedures.
- Relevant information is distributed across multiple channels:
  - Moodle
  - Mathplan timetable system
  - university email activation
  - orientation week procedures
  - semester ticket
  - examination systems
  - administrative tasks in Germany
- New students often do not know which source is authoritative for a specific question.
- As a result, onboarding becomes slower and more confusing than necessary.

Motivation: Information Fragmentation



Information fragmentation during onboarding

Current Problem

- Students repeatedly ask operational questions during the first weeks of the semester.
- Answers already exist, but they are spread across different documents, portals, and contact points.
- This causes repeated explanations and slows down onboarding for both students and staff.

Current Problem: Typical Questions and Sources

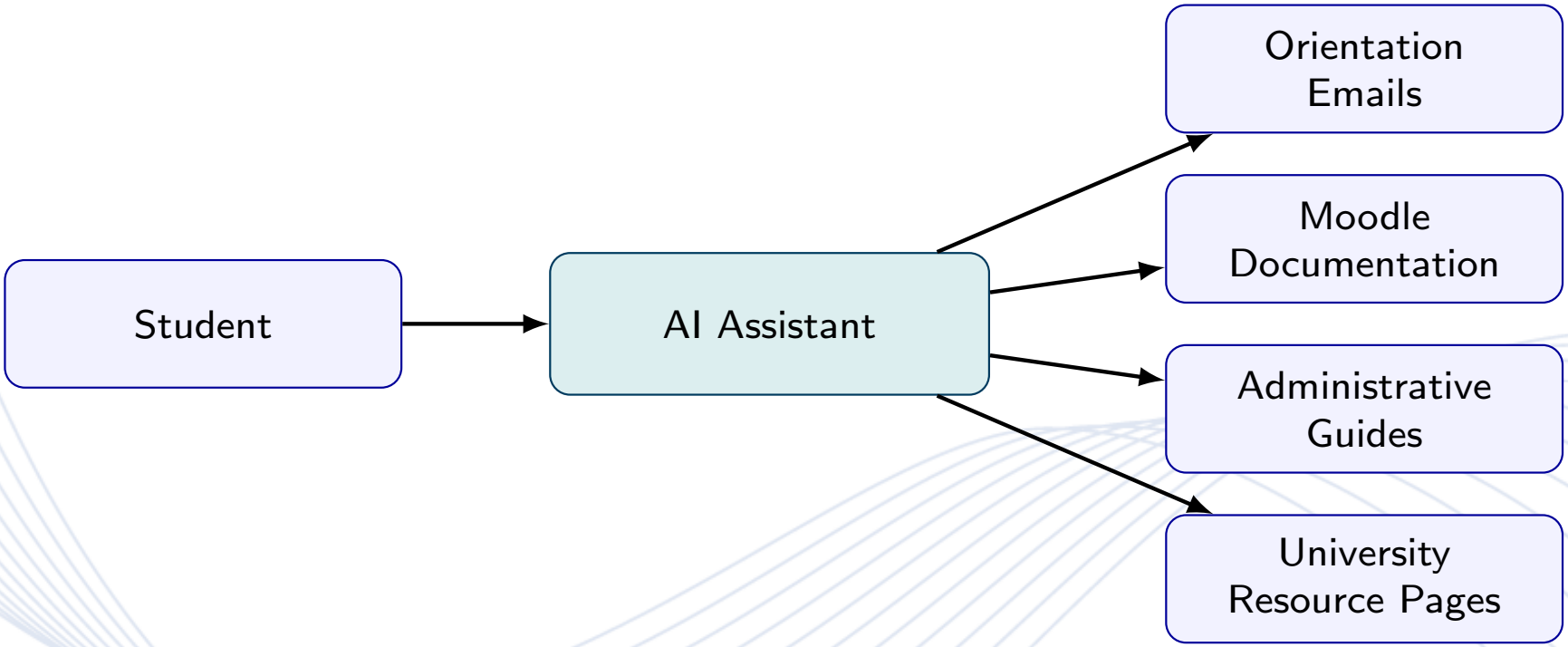
Category	Example Question	Current Source
University Account	How do I activate email?	Orientation email
Course Systems	How do I access Moodle?	IT documentation
Timetable	How do I use Mathplan?	Moodle guide
Administration	How do I get semester ticket?	University portal
Exams	Where can I find transcript?	Examination office

Concept Proposal

Proposed Solution

- Concept: a **Local LLM-based Student Support Assistant**.
- The assistant should:
  - answer student questions
  - retrieve information from official documents
  - guide users through university processes
  - provide relevant links to resources
  - cite the underlying source documents
- The objective is not to replace staff, but to provide a first, structured support layer.

Proposed Solution: Knowledge Hub



Knowledge hub concept

Example Interaction

Example question

Student Input

How do I activate my university email?

Expected response structure

- step-by-step instructions
- link to the activation portal
- explanation of required information
- citation of the source document

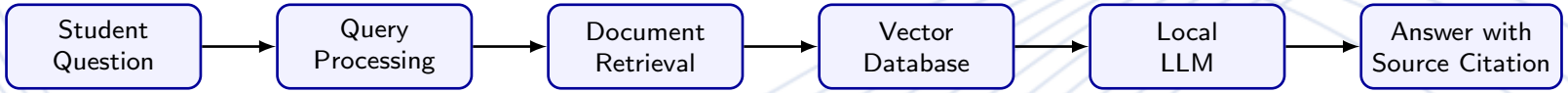
User question:
How do I activate my university email?

Assistant response:
1. Open activation portal
2. Enter student credentials
3. Set password
4. Confirm mailbox access

Source reference:
Orientation email + IT activation guide

System Concept

- The system follows a retrieval-based architecture instead of relying only on model memory.
- The assistant first searches relevant university material, then generates an answer grounded in those sources.
- This improves traceability and supports source citation in responses.

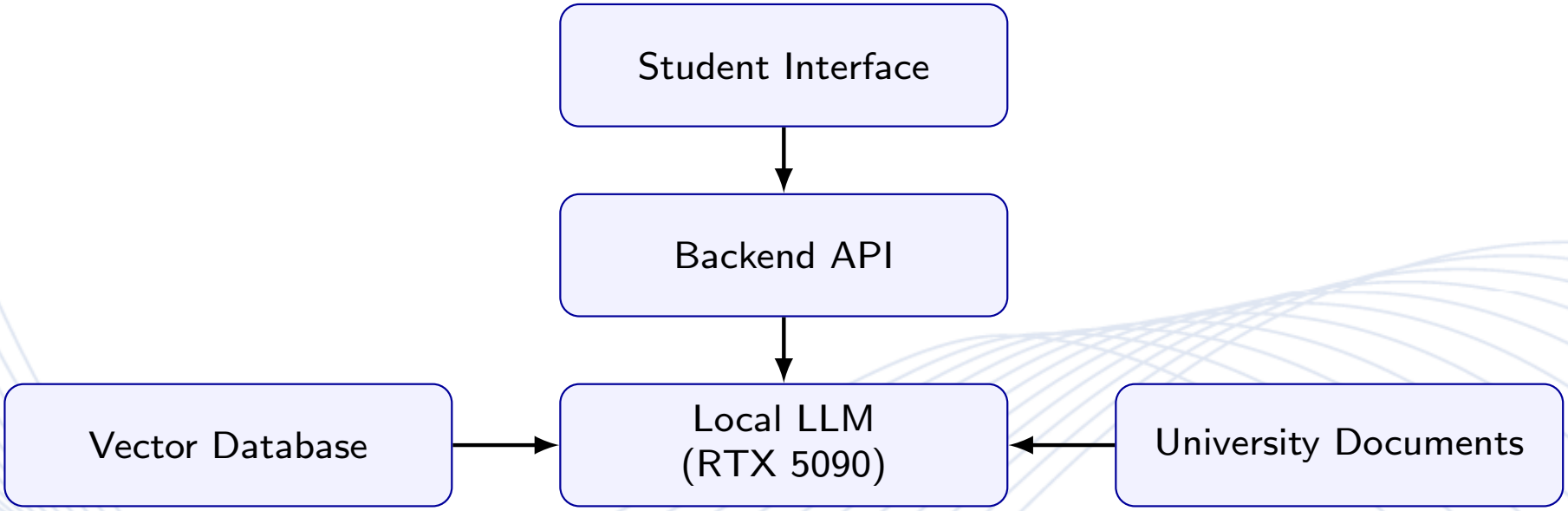


Prototype and Deployment

Prototype Implementation

- The prototype is planned as a **local system** running on the **RTX 5090 GPU available in the lab**.
- This setup is suitable to demonstrate:
  - local inference
  - privacy-preserving architecture
  - effective use of available GPU resources
- The goal is a proof of concept for student support, not a production deployment.

Prototype Implementation: Local Architecture



Possible Deployment (Future Concept)

Access Method	Description	Comments
Web Application	Browser-based chat interface accessible from any device	Can be linked from Moodle or shared via QR code
Messaging Bot	Chat interface via Telegram or WhatsApp	Familiar interface; mobile-friendly access
Internal Web App (Campus Network)	Web interface accessible only within university network	Can integrate with Moodle or internal systems

Expected Value and Feedback

Potential Benefits

For Students

- Faster answers to common questions (timetable, Moodle, email activation, semester ticket)
- Centralized access to important study information
- 24/7 access to guidance for administrative processes (potential feature)

For Coordination Office

- Reduced workload from repetitive onboarding queries
- Support available even when staff are not immediately reachable
- Better focus on complex student issues rather than routine guidance

For University

- Potential foundation for a university-wide digital student assistant
- Scalable architecture that could support other programs and departments

Questions and Feedback

1. **General feedback on the idea**
   Could this concept work as a project for **Project B under the "Local LLM Agent" topic**? We would appreciate any feedback on whether this idea is appropriate for the course.
2. **Scope of the Local LLM Agent**
   How broad should the scope of the assistant be? Should we focus specifically on the **MBIDA + TM programs**, or consider a more general assistant for **international students at the university**?

Questions and Feedback (cont.)

3. **Type of prototype**
   What type of prototype would you prefer to see for this project?
   - A **local prototype running on the RTX 5090 or a laptop**, mainly demonstrating the AI architecture and retrieval system
   - Or a **simple web application** that students could potentially start using next semester
4. **Research directions and resources**
   Are there any **technical directions, research aspects, or resources** you would recommend exploring for this project?

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Thanks a lot

for your attention